BudgetBuddy - Intelligent Personal Finance Assistant

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1. Problem Statement and Target Users

Managing personal finances is complex and overwhelming for many individuals. Users often struggle to track spending across multiple sources, interpret credit card statements, plan savings or debt repayment, and adjust budgets based on real-world factors such as inflation and cost of living. Existing budgeting tools focus heavily on manual entry and static dashboards, offering limited personalized insights or conversational guidance.

Primary Users: Individuals aged 18-45, tech-savvy, seeking to understand and improve their financial situation

- Young professionals managing their first significant income
- Self-employed individuals with variable income
- Couples/families coordinating finances
- Parents wanting to teach their children financial literacy
- Anyone wanting to optimize savings and reach financial goals

Secondary Users: Financial educators, financial wellness programs in companies

2. Proposed Solution and Key Features

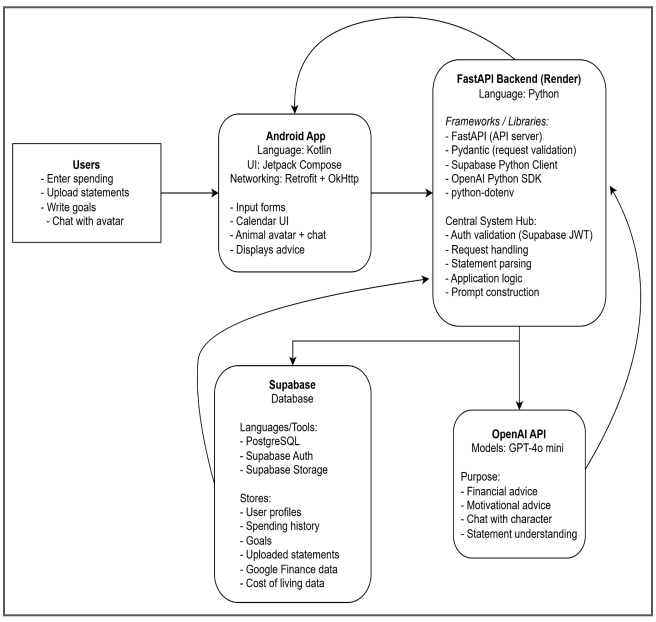
BudgetBuddy is an AI-powered personal finance assistant that combines structured financial data with Large Language Models (LLMs) to deliver personalized budgeting advice, reminders, and financial insights through a conversational interface.
The system allows users to input income, spending habits, and financial goals. It augments this data with external financial and cost-of-living information, then uses LLMs to generate intelligent recommendations.

Features:
💵 **Automated Statement Parsing:** Users can upload credit card statements and expense receipts for automatic parsing and categorization.
📈 **Goal Setting & Tracking:** Users set financial goals and track progress.
📊 **Cost-of-Living & Financial Data Integration:** Real-time data from external APIs for personalized insights.
🤖 **Conversational Agent:** An animal character provides chat-based financial advice and reminders.
🎯 **Smart Features:**

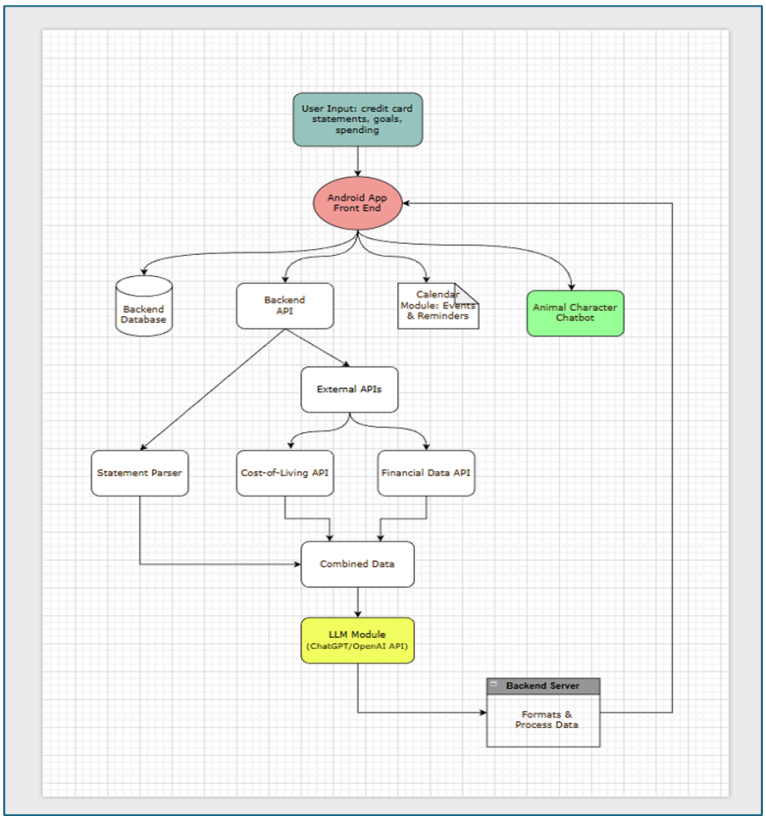
- **Calendar Integration:** Events and reminders for bill payments and financial milestones.
- **Multimodal Capabilities:** Image upload/vision for receipts and statements.
- **CRUD Operations:** Manage income, expenses, and goals.
- **Real-Time Metrics:** Visual dashboards and analytics.
- **Agent-Based Recommendations:** Personalized suggestions using LLMs.

3. Technical Approach and Stack

- **Frontend:** Android App (Kotlin/Jetpack Compose)
- **Backend Orchestration:** Python (FastAPI)
- **LLM Support:** OpenAI GPT-4o
- **Database:** Supabase, PostgreSQL (for user data, transactions, goals)
- **Vector Store:** Supabase, pgvector extension (for RAG and semantic search)
- **APIs:**OpenAI API
- **Deployment:** Render



4. System Architecture Diagram



5. Success criteria and evaluation metrics

Functional Requirements

✔ All pages work without errors
✔ CRUD operations persist across sessions
✔ API requests complete
✔ Graceful fallback when API unavailable

Performance Metrics

- **Accuracy:** Correct parsing and categorization of statements (target: >75%).
- **Response Quality:** User ratings of LLM-generated advice (target: >3/5).
- **System Reliability:** Uptime and error rates (target: >60% uptime).
- **Deployment Readiness:** Successful deployment